# Collision in a circle

Y16 (2016) — English translation

On the air hockey table there are $N$ identical small pucks, evenly arranged in a semicircle (see picture); the total mass of the washers is $M$. Another small washer, disc $D$, of mass $m$, slides in a direction perpendicular to the diameter of this semicircle and collides with the first of the stationary washers. By some miracle it happened that later it collided with all the other $N-1$ washers in order, after which it continued to move in the direction opposite to the initial one. All collisions are absolutely elastic; friction can be neglected everywhere.

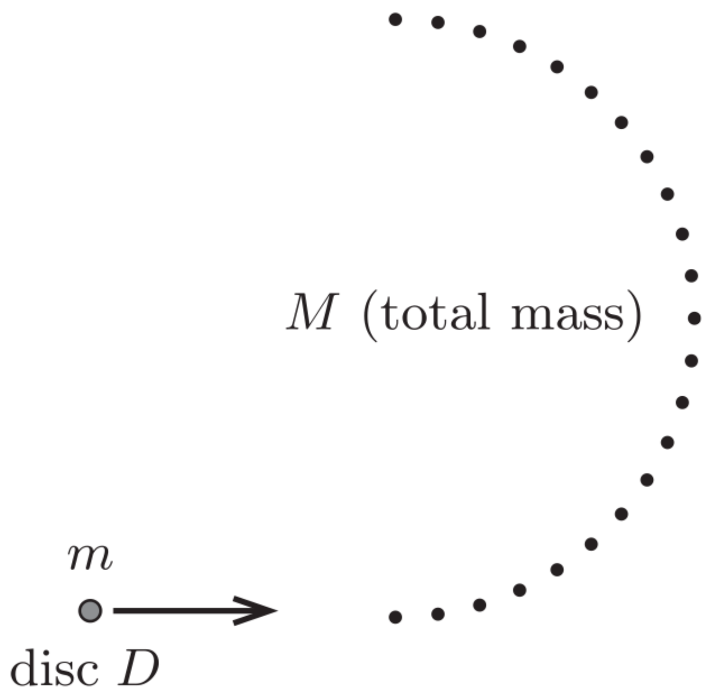


**Figure 1:** Figure 1.

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|---|---|---|
| **a** | In the limiting case at $N \to \infty$, what is the minimum value of the ratio $M/m$ at which the above miracle is possible? | **0** |

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|---|---|---|
| **b** | If the mass ratio is equal to the critical value found in part $a$), what is the ratio of the final and initial speed $D$? | **0** |

*Source:* `https://pho.rs/p/2220`